



UNIVERSITY OF CAPE TOWN

CSC200SS: Computer Science

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1 Parallel and Concurrent Programming

2 Theory of Computation

2.1 Finite automata

Definition: Finite automata

A **finite automaton** is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where:

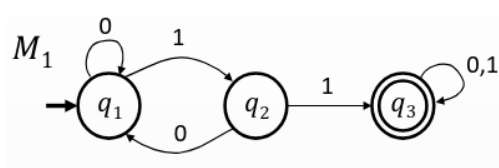
- Q is a finite set of all possible states.
- Σ is a finite set of all alphabet symbols.
- δ is a transition function $\delta : Q \times \Sigma \rightarrow Q$. In other words,

$$\delta(q, a) = r \text{ means } \textcircled{q} \xrightarrow{a} \textcircled{r}$$

- q_0 is the start state.
- F is the set of accept states.

Example:

Let's consider the following finite automaton.



Here, $M_1 = (Q, \Sigma, \delta, q_1, F)$ is a finite automaton. For sake of example, here are the exact details of M_1 .

- $Q = \{q_1, q_2, q_3\}$
- $\Sigma = \{0, 1\}$
- $F = \{q_3\}$

It accepts a finite string as input. The output is either *ACCEPT* or *REJECT*. The computation goes as follows:

1. Begin at start state.
2. Read input symbols.
3. Follow corresponding transitions.
4. *ACCEPT* if we reach the accept state, otherwise *REJECT*.

For example: $01101 \rightarrow \text{ACCEPT}$, while $00101 \rightarrow \text{REJECT}$. Here, M_1 accepts all string in A where $A = \{w | w \text{ contains substring } 11\}$. We say that A is the language of $M_1 \iff M_1$ recognizes $A \iff A = L(M_1)$.

2.2 Strings and languages

A **string** is a finite sequence of symbols $w_1w_2\cdots w_n \in \Sigma$. A **language** is a set of string (finite or infinite). The **empty string** ϵ is the string of length 0. The **empty language** $\emptyset$ is the set with no strings.

Definition: Accepting strings

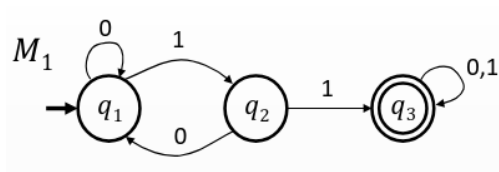
A finite automaton M **accepts string** $w = w_1w_2\cdots w_n$ for each $w_i \in \Sigma$ if there exists a sequence of states $r_0, r_1, \dots, r_n \in Q$ where:

- $r_0 = q_0$
- $r_i = \delta(r_{i-1}, w_i)$ for $1 \leq i \leq n$
- $r_n \in F$

Definition: Regular language

We say a language is **regular** if some finite automaton recognizes it.

Example 1:



$L = (M_1) = \{w | w \text{ contains substring } 11\} = A$. Therefore, A is regular.

Example 2:

Let $B = \{w | w \text{ has an even number of 1s}\}$. In this case, B is regular. (not shown)

Example 3:

Let $C = \{w | w \text{ has equal number of 0s and 1s}\}$. In this case, C is not regular. (provable)

2.3 Regular expressions

Definition: Regular operations

Let A, B be languages.

- **Union:** $A \cup B = \{w | w \in A \text{ or } w \in B\}$.
- **Concatenation:** $A \circ B = \{xy | x \in A \text{ and } y \in B\} = AB$.
- **Star:** $A^* = \{x_1 \dots x_k | x_i \in A \text{ for } k \geq 0\}$. (Note: $\epsilon \in A^*$)

Regular expressions are built from $\Sigma, \emptyset, \epsilon$ by using regular operations $\cup, \circ, \star$.

Example 1:

- $(0 \cup 1)^* = \Sigma^*$ gives all strings over Σ .
- Σ^*1 gives all strings that end with 1.
- $\Sigma^*11\Sigma^*$ gives all strings that contain 11 = $L(M_1)$.

Example 2:

Let $A = \{\text{good, bad}\}$ and $B = \{\text{boy, girl}\}$.

- $A \cup B = \{\text{good, bad, boy, girl}\}$.
- $A \circ B = AB = \{\text{goodboy, goodgirl, badboy, badgirl}\}$.
- $A^* = \{\epsilon, \text{good, bad, goodgood, goodbad, badgood, badbad, goodgoodgood, } \dots\}$